

# LAB NOTES: PDF GENERATION EXPERIMENT

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**CLASSIFICATION:** FOR OFFICIAL USE ONLY

## Lab Notes: PDF Generation Experiment

**Researcher:** Dr. WAFT

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**Experiment ID:** EXP-2026-01-13-001

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### Hypothesis

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PDF generation can be automated using evolutionary algorithms that adapt content selection and styling based on target constraints.

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# Materials

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- Python 3.12
  - WeasyPrint 67.0
  - WAFT PDF Generation Framework
  - Content samples (15+ documents)
  - Styling genomes (clinical\_standard, premium, professional)
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# Procedure

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## 1. **Content Preparation**

2. Collected diverse content samples
3. Marked content with metadata
4. Prepared for distillation

## 5. **Distillation Phase**

6. Used ChatDistiller to extract ideas
7. Ranked ideas by importance
8. Grouped related ideas

## 9. Styling Phase

10. Selected styling genome

11. Applied preset configurations

12. Customized as needed

## 13. Generation Phase

14. Generated HTML from template

15. Converted to PDF using WeasyPrint

16. Validated output

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# Observations

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## Trial 1: Poetry Collection

- **Result:** Success
- **Pages:** 3
- **Quality:** High
- **Notes:** Premium style worked well for creative content

## Trial 2: Technical Manifesto

- **Result:** Success
- **Pages:** 2
- **Quality:** High
- **Notes:** Clinical standard style appropriate for technical content

## Trial 3: Scientific Paper

- **Result:** Success
- **Pages:** 2 (target achieved)
- **Quality:** High
- **Notes:** Two-page constraint successfully enforced

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## Analysis

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The evolutionary PDF generation system successfully:

1. Adapted content selection to page constraints
  2. Applied appropriate styling based on content type
  3. Maintained quality across diverse document types
  4. Enforced constraints (e.g., 2-page limit) when required
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## Conclusion

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The hypothesis is **CONFIRMED**. Evolutionary algorithms can successfully automate PDF generation with adaptive content selection and styling.

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## Next Steps

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1. Test with larger content sets
  2. Explore additional styling genomes
  3. Implement feedback loop for quality improvement
  4. Document best practices for different content types
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**End of Experiment**